A QUADRATIC QUADRILLE IN PARABOLIC PANTALOONS

Abstract

Some observations on the effects of extending a quadratic equation to the field of complex numbers. A particular focus is on the location and behaviour of the vertex and roots of the quadratic's modulus under the influence of complex parameters. While there is a great deal to explore in such an object, much of the following effort is devoted to finding an explicit expression of its roots.

Project 2
Mathematics 2130

Submitted by: Jonathan Kirkpatrick
Submitted to: Ivan Booth

December 17, 2025

1 Introduction

The quadratic is one of the first functions many students encounter. It seems only fitting it should provide an introduction to the behaviour of functions over the field of complex numbers as well. Specifically, its absolute value, or modulus, will form the foundation of this brief exposition. The function's plot is a marginally exotic form; it appears to be a pair of trousers with the legs sewn shut in lieu of cuffs, and standing on the xy plane. Where the legs make contact with the plane, we will call roots. Maintaining our sartorial theme, the top of its inseam will be the vertex. We will break the function apart to get a handle on its intricacies before observing the effects of its complex variables and coefficients on its form and position. By approaching the problem in this order, we maximize our potential for insight.

2 Wrangling an Unruly Function

Coming to terms with a new object, it's critical to gain an understanding of its defining features. This seems a good time to introduce the complex quadratic's modulus in symbolic form. It emerges naturally by replacing our real variables and coefficients with complex ones. The standard form of a quadratic is

$$y = Ax^2 + Bx + C,$$

where each of the capital letters represents a real number along with the variable, x. If we aren't interested in plotting this on the Cartesian plane, we may just want to know how far x is from y; for this, we can take the absolute value of the function

$$y = |Ax^2 + Bx + C| = \sqrt{(Ax^2 + Bx + C)^2}.$$

As a bit of an aside, when we make the switch to complex numbers, I don't feel great about using the term absolute value, so I will use *modulus* going forward. This is meant to represent the same idea. The imaginary portion of the coefficients are represented with Greek letters and the real with Latin ones. We will adopt a convention to use x for real and y for imaginary with respect to the complex variable. The result of the function will be plotted on the z axis. Below is our familiar quadratic function, over the field of complex numbers

$$g(x, y) = (a + i\alpha)(x + iy)^2 + (b + i\beta)(x + iy) + c + i\gamma.$$

In this form, we have a bit of an alphabet soup of an equation. Before we make sense of it, we will expand it out entirely and partition it into real and imaginary components

$$ax^2 + i\,\alpha x^2 + 2i\,axy - 2\,\alpha xy - ay^2 - i\,\alpha y^2 + bx + i\,\beta x + i\,by - \beta y + c + i\,\gamma.$$

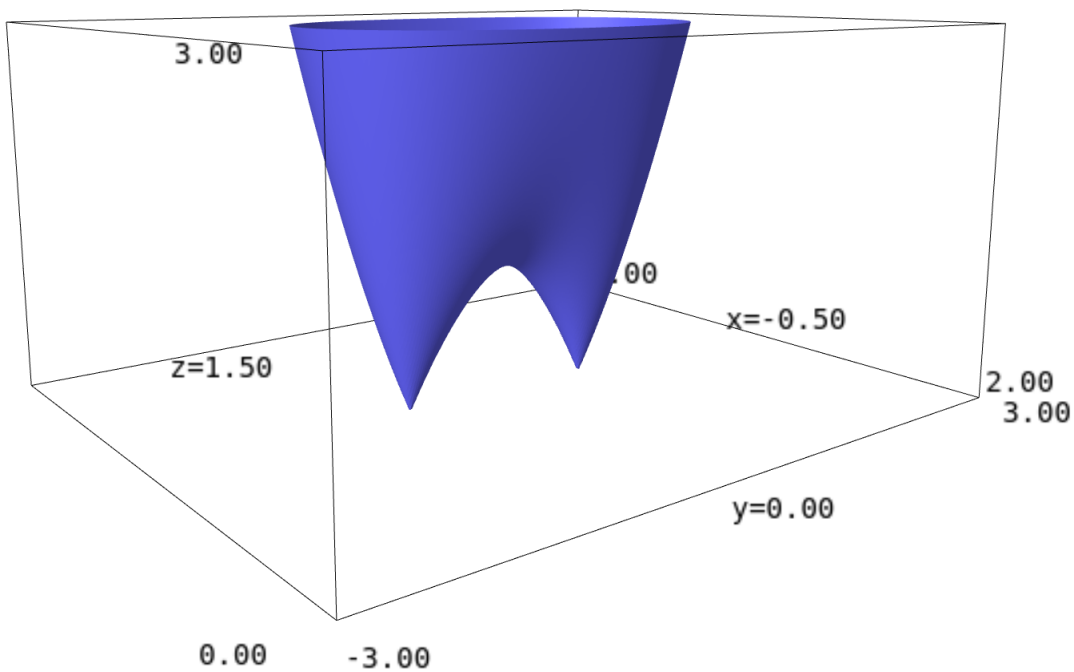
Now we have it all expanded, and can partition it as

$$(ax^2 - 2\,\alpha xy - ay^2 + bx - \beta y + c) + i(\alpha x^2 + 2\,axy - \alpha y^2 + \beta x + by + \gamma).$$

These are the real and complex portions that will make up our modulus. We have what appears to be a pair of conics representing the real and imaginary portions of our quadratic. We would like to know how far a point lies from the origin in the complex plane, given a value of x and y. This value will be recorded in the z-axis courtesy of Pythagoras. Now in its final form, we have the function that will define our quadratic modulus over the field of complex numbers

$$f(x, y) = \sqrt{(ax^2 - 2\,\alpha xy - ay^2 + bx - \beta y + c)^2 + (\alpha x^2 + 2\,axy - \alpha y^2 + \beta x + by + \gamma)^2} \tag{1}$$

Having found a representation of the modulus, we will turn our attention to teasing out some of its critical points to grant a better frame of reference when we begin our analysis of the quadratic. Below is a plot of the modulus, taking all coefficients to have a value of 1; note the trousine form of the plot.



With the equation now defined and visual representation available, a logical next step is to determine the vertex.

2.1 The Vertex

While complex numbers obey the same rules as real numbers from an algebraic standpoint, they do introduce significant impediments to manipulation if we would like to rid our expressions of the imaginary unit, i. Fortunately, the location of the vertex transfers cleanly into a complex quadratic from its real cousin. For this, we need only replace the real coefficients with complex ones and subsequently split it into real and imaginary portions.

$$-\frac{B}{2A} \to -\frac{b+i\beta}{2a+i2\alpha} \to \left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right) + i\left(\frac{\alpha b - a\beta}{2a^2+2\alpha^2}\right)$$

Moving forward, we will denote the coordinates of the vertices as:

$$x_{\text{vertex}} = \frac{ab+\alpha\beta}{2a^2+2\alpha^2} \tag{2}$$

$$y_{\text{vertex}} = \frac{\alpha b - a\beta}{2a^2+2\alpha^2} \tag{3}$$

$$z_{\text{vertex}} = f(x_{\text{vertex}}, y_{\text{vertex}}) \tag{4}$$

Later in our work, these coordinates will allow us to simplify the location of the roots considerably.

2.2 The Intersection of Conics

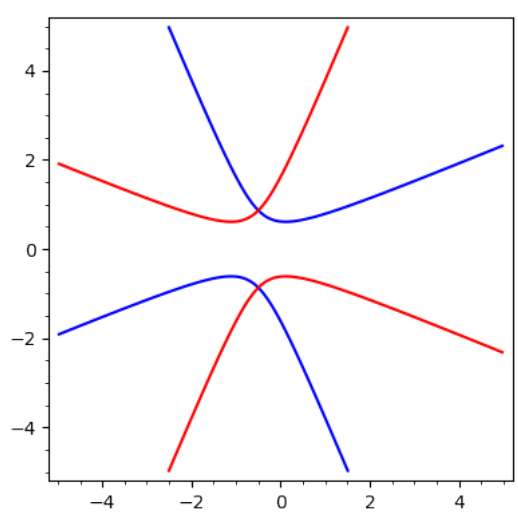
SageMath is a powerful tool to analyze and manipulate expressions; coupled with its builtin python environment, it can make short work of many complicated tasks. Despite its many sterling qualities, there are some limitations in the hands of a novice such as myself. Taking a naive approach to root finding with our complex quadratic, we arrive at the rather unsatisfactory

$$x = \frac{a\sqrt{|b^2 + 2i\,b\beta - \beta^2 - 4\,ac - 4i\,\alpha c - 4i\,a\gamma + 4\,\alpha\gamma|}\cos\left(\frac{1}{2}\arctan\left(2\,b\beta - 4\,\alpha c - 4\,a\gamma, b^2 - \beta^2 - 4\,ac + 4\,\alpha\gamma\right)\right)}{2\left(a^2 + \alpha^2\right)}$$
$$+ \frac{\alpha\sqrt{|b^2 + 2i\,b\beta - \beta^2 - 4\,ac - 4i\,\alpha c - 4i\,a\gamma + 4\,\alpha\gamma|}\sin\left(\frac{1}{2}\arctan\left(2\,b\beta - 4\,\alpha c - 4\,a\gamma, b^2 - \beta^2 - 4\,ac + 4\,\alpha\gamma\right)\right)}{2\left(a^2 + \alpha^2\right)} - b.$$

This may in fact represent the precise location of the x coordinate of one of the roots of the equation. I don't propose we test this theory. Rather, we can use a few facts about the location of potential roots and the behaviour of conics to guide our exploration. We will first observe from (1), that in order to have a root, both the real and imaginary portion must equal zero. If we set each to zero, we have a pair of conics in the xy-plane representing where the real and imaginary conics respectively evaluate to zero. Where the two conics intersect, we have our roots. A second observation about the equations in question is that any linear combination of them will alter the form of the conic, but preserve incidence. Since it is precisely incidence we are interested in, we will transform the equations into a form that will better enable us to tease out an expression for the roots. Here are the real and imaginary parts of our quadratic expressed in the form of their respective conics:

$$0 = ax^2 - 2\alpha xy - ay^2 + bx - \beta y + c \tag{5}$$
$$0 = \alpha x^2 + 2axy - \alpha y^2 + \beta x + by + \gamma \tag{6}$$

As seen below, these two conics are hyperbolas and their major axis are not trivial. Once again, we have set the value of all parameters to one. The blue plot is the real conic and the red is the imaginary.



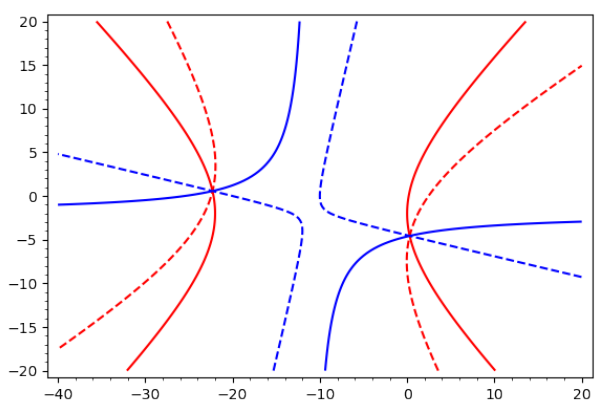
In their present form, the influence of the six parameters is too complicated to hope for a reasonable expression of the conics' intersection. Where we may achieve some success is by decoupling the parameters from the variables. Since we are able to change the hyperbolas while maintaining incidence by adding or subtracting multiples of the equations to each other, we will seek a particular

form of value. We would like to remove the x^2 and y^2 from one and the xy term from the other. This will result in a simplified version of each, and more susceptible to further manipulation.

$$\begin{aligned}
&ax^2 - 2\alpha xy - ay^2 + bx - \beta y + c \\
&+ (\alpha x^2 + 2axy - \alpha y^2 + \beta x + by + \gamma)\left(\frac{\alpha}{a}\right) \\
&= \frac{\alpha^2 + a^2}{a}(x^2 - y^2) + \frac{\alpha\beta + ab}{a}x + \frac{\alpha b - a\beta}{a}y + \frac{ac + \alpha\gamma}{a} \\
&= (\alpha^2 + a^2)(x^2 - y^2) + (\alpha\beta + ab)x + (\alpha b - a\beta)y + (ac + \alpha\gamma)
\end{aligned} \tag{7}$$

$$\begin{aligned}
&\alpha x^2 + 2axy - \alpha y^2 + \beta x + by + \gamma \\
&- (ax^2 - 2\alpha xy - ay^2 + bx - \beta y + c)\left(\frac{\alpha}{a}\right) \\
&= \frac{2a^2 + 2\alpha^2}{a}xy + \frac{a\beta - \alpha b}{a}x + \frac{ab + \alpha\beta}{a}y + \frac{a\gamma - \alpha c}{a} \\
&= (2a^2 + 2\alpha^2)xy + (a\beta - \alpha b)x + (ab + \alpha\beta)y + (a\gamma - \alpha c)
\end{aligned} \tag{8}$$

We have now transformed the two conics into forms that appear slightly more unwieldy at first, but are far more civilized in the xy plane. In their original form, modification to any of their parameters except c and γ would result in a rotation of their major axis; in their new guise the axis is fixed with respect to the vertex under the action of all their parameters. Below is a plot of the new conics along with their forebears. The original conics are present as dashed lines. I have adjusted the values of the parameters to make the behaviour change of our new conics more pronounced.



We observe that the blue conic has both of its axis at $\pm 45°$ centred on the vertex while the red one has a vertical and horizontal axis again centred on the vertex. The axis of the original conics are centred on the vertex as well, but far more exotic in their orientation.

Taking our conics one at a time, we are now in a position to manipulate their equations to find a representation in a vastly more compact and useful form. We will begin with the conic involving x^2 and y^2. Specifically, we will start by only considering terms involving x^2.

$$\begin{aligned}
(a^2+\alpha^2)x^2+(ab+\alpha\beta)x &= (a^2+\alpha^2)\left(x^2+\frac{ab+\alpha\beta}{a^2+\alpha^2}x\right)\\
&= (a^2+\alpha^2)\left(x^2+\frac{ab+\alpha\beta}{a^2+\alpha^2}x+\left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2-\left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2\right)\\
&= (a^2+\alpha^2)\left(\left(x+\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2-\left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2\right)\\
&= (a^2+\alpha^2)\left((x+x_{\text{vertex}})^2-x_{\text{vertex}}^2\right) \qquad (9)
\end{aligned}$$

The major note here is the emergence of x_{vertex} in our expression. We will continue by completing the square with the terms involving y^2.

$$\begin{aligned}
(\alpha b-a\beta)y-(\alpha^2+a^2)y^2 &= -(a^2+\alpha^2)\left(y^2-\frac{\alpha b-a\beta}{a^2+\alpha^2}y\right)\\
&= -(a^2+\alpha^2)\left(y^2-\frac{\alpha b-a\beta}{a^2+\alpha^2}y+\left(\frac{\alpha b-a\beta}{2(a^2+\alpha^2)}\right)^2-\left(\frac{\alpha b-a\beta}{2(a^2+\alpha^2)}\right)^2\right)\\
&= -(a^2+\alpha^2)\left((y+y_{\text{vertex}})^2-y_{\text{vertex}}^2\right) \qquad (10)
\end{aligned}$$

With the squares completed for both the x and y components of the conic, we can now reassemble it to yield:

$$\begin{aligned}
0 &= (a^2+\alpha^2)\left(\left((x+x_{\text{vertex}})^2-x_{\text{vertex}}^2\right)-\left((y+y_{\text{vertex}})^2-y_{\text{vertex}}^2\right)\right)+(ac+\alpha\gamma)\\
&= \left((x+x_{\text{vertex}})^2-x_{\text{vertex}}^2\right)-\left((y+y_{\text{vertex}})^2-y_{\text{vertex}}^2\right)+\frac{ac+\alpha\gamma}{a^2+\alpha^2}\\
&= (x+x_{\text{vertex}})^2-(y+y_{\text{vertex}})^2-\left(x_{\text{vertex}}^2-y_{\text{vertex}}^2-\frac{ac+\alpha\gamma}{a^2+\alpha^2}\right) \qquad (11)
\end{aligned}$$

Before moving to the other conic, we will combine our constant terms into one,

$$p = x_{\text{vertex}}^2-y_{\text{vertex}}^2-\frac{ac+\alpha\gamma}{a^2+\alpha^2}$$

and shift the conic to the origin

$$0 = x^2-y^2-p. \qquad (12)$$

This is a far more pleasant representation of what originally seemed like a terribly complex equation. Some of the ugliness is hidden from view in the constant term, p, but it is just a constant term after all and the overall complexity has been reduced significantly. The second conic can receive a similar treatment as follows:

$$\begin{aligned}
0 &= (2a^2+2\alpha^2)xy+(a\beta-\alpha b)x+(ab+\alpha\beta)y+(a\gamma-\alpha c)\\
&= xy+y_{\text{vertex}}x+x_{\text{vertex}}y+\frac{a\gamma-\alpha c}{2a^2+2\alpha^2}
\end{aligned}$$

Prior to further simplification, we will shift the conic to the origin and expand all the terms.

$$\begin{aligned}0 &= (x - x_{\text{vertex}})(y - y_{\text{vertex}}) + y_{\text{vertex}}(x - x_{\text{vertex}}) + x_{\text{vertex}}(y - y_{\text{vertex}}) + \frac{a\gamma - \alpha c}{2a^2 + 2\alpha^2} \\ &= xy - x_{\text{vertex}}y_{\text{vertex}} + \frac{a\gamma - \alpha c}{2a^2 + 2\alpha^2}\end{aligned}$$

As with the other equation, we will collect our constants into one

$$q = x_{\text{vertex}}y_{\text{vertex}} - \frac{a\gamma - \alpha c}{2a^2 + 2\alpha^2}$$

so our final equation for this conic is

$$0 = xy - q. \tag{13}$$

With the problem reduced to something tractable, we can finally determine a general expression for the x and y components of the roots in terms of the parameters alone. We will begin by solving for x and substituting the expression into the other equation.

$$x = \frac{q}{y}$$

$$x^2 - y^2 - p = \left(\frac{q}{y}\right)^2 - y^2 - p = \frac{q^2}{y^2} - y^2 - p = 0 \rightarrow \boxed{y^4 + py^2 - q = 0}$$

We have a quadratic in y^2 and can employ the quadratic formula to solve for its roots and subsequently take the square root of that to express the roots of y

$$y = -\sqrt{\frac{-p \pm \sqrt{p^2 + 4q^2}}{2}}.$$

With a solution to x above, we can perform our final substitution to yield an expression for the roots of x

$$x = -\frac{q}{\sqrt{\frac{-p \pm \sqrt{p^2+4q^2}}{2}}}.$$

Much of the intricacy of the equations is hidden within the constants, and thankfully this is the case. Below is a partial expansion of the explicit formula for the x and y values for one of its roots.

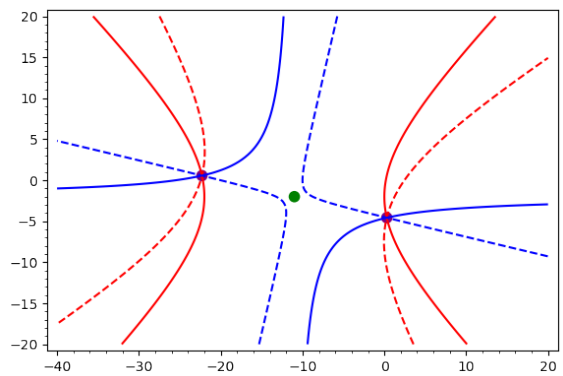
$$\begin{aligned}y &= \sqrt{\frac{-p \pm \sqrt{p^2 + 4q^2}}{2}} \\ &= \sqrt{\frac{-(x_{\text{vertex}}^2 - y_{\text{vertex}}^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2}) \pm \sqrt{(x_{\text{vertex}}^2 - y_{\text{vertex}}^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2})^2 + 4(x_{\text{vertex}}y_{\text{vertex}} - \frac{a\gamma-\alpha c}{2a^2+2\alpha^2})^2}}{2}} \\ &= \sqrt{\frac{-\left(\left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2 - (\frac{\alpha b-a\beta}{2a^2+2\alpha^2})^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2}\right) \pm \sqrt{((\frac{ab+\alpha\beta}{2a^2+2\alpha^2})^2 - (\frac{\alpha b-a\beta}{2a^2+2\alpha^2})^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2})^2 + 4((\frac{ab+\alpha\beta}{2a^2+2\alpha^2})(\frac{\alpha b-a\beta}{2a^2+2\alpha^2}) - \frac{a\gamma-\alpha c}{2a^2+2\alpha^2})^2}}{2}}\end{aligned}$$

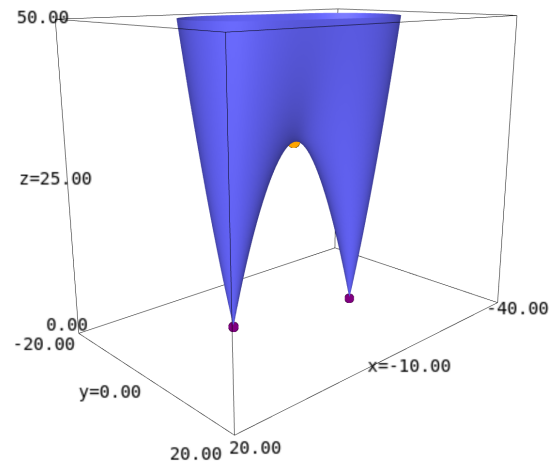
$$x = \frac{(\frac{ab+\alpha\beta}{2a^2+2\alpha^2})(\frac{\alpha b-a\beta}{2a^2+2\alpha^2}) - \frac{a\gamma-\alpha c}{2a^2+2\alpha^2}}{\sqrt{\frac{-\left(\left(\frac{ab+\alpha\beta}{2a^2+2\alpha^2}\right)^2 - (\frac{\alpha b-a\beta}{2a^2+2\alpha^2})^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2}\right) \pm \sqrt{((\frac{ab+\alpha\beta}{2a^2+2\alpha^2})^2 - (\frac{\alpha b-a\beta}{2a^2+2\alpha^2})^2 - \frac{ac+\alpha\gamma}{a^2+\alpha^2})^2 + 4((\frac{ab+\alpha\beta}{2a^2+2\alpha^2})(\frac{\alpha b-a\beta}{2a^2+2\alpha^2}) - \frac{a\gamma-\alpha c}{2a^2+2\alpha^2})^2}}{2}}}$$

Bear in mind that these are the roots that occur over the origin. They will still need to be shifted back to the vertex before they can be of use. Given the enormity of the equations, we will revert to using their abbreviated form, shifted to the vertex and refer to them as

$$x_{r1},\ x_{r2},\ y_{r1},\ \text{and}\ y_{r2}$$

in subsequent sections. We can now place them in our plots as we explore the impacts the parameters have on the form of the modulus.

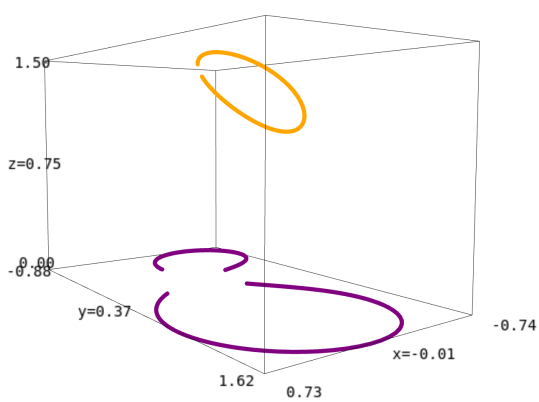


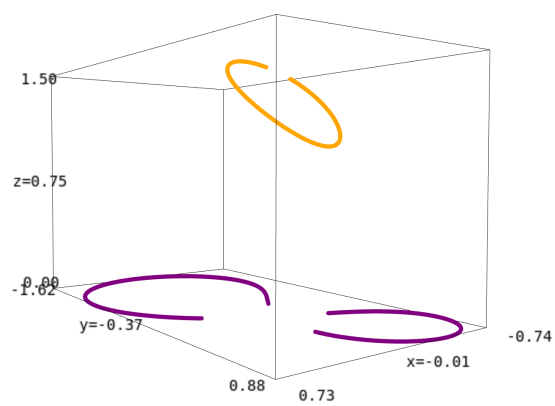


3 Choreography in Six Parameters

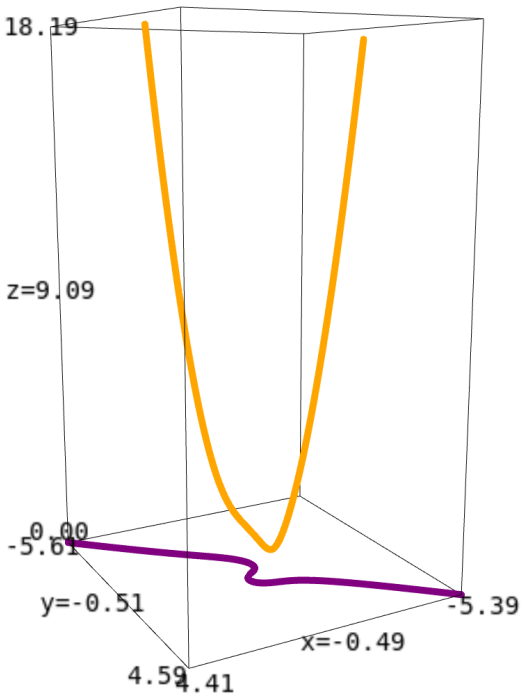
Given the complex interplay of our coefficients, learning about their effects will best be achieved by a graphical rather than symbolic representation. With explicit equations for each of the five critical values, we will observe their behaviour by parametrization. As mentioned earlier, our primary focus is on the xy plane, but in order to include the vertex we will present 3d plots. Below is a series of plots showing the effect each coefficient has on the location of the roots and vertex.

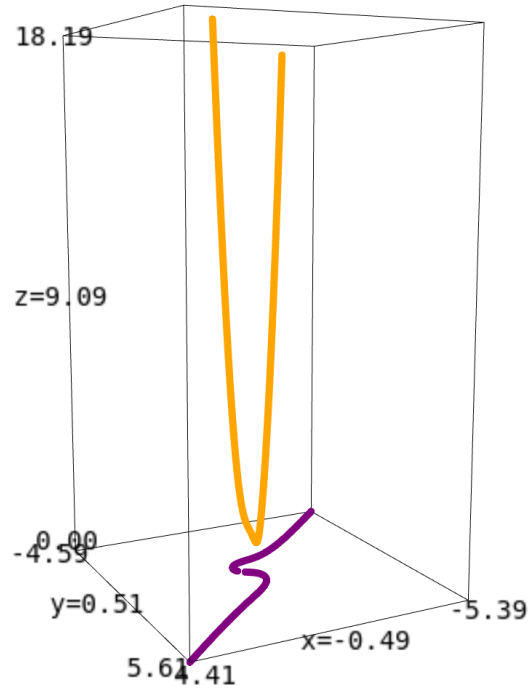
The first two figures represent the result of varying a and α between -10 and 10 respectively.



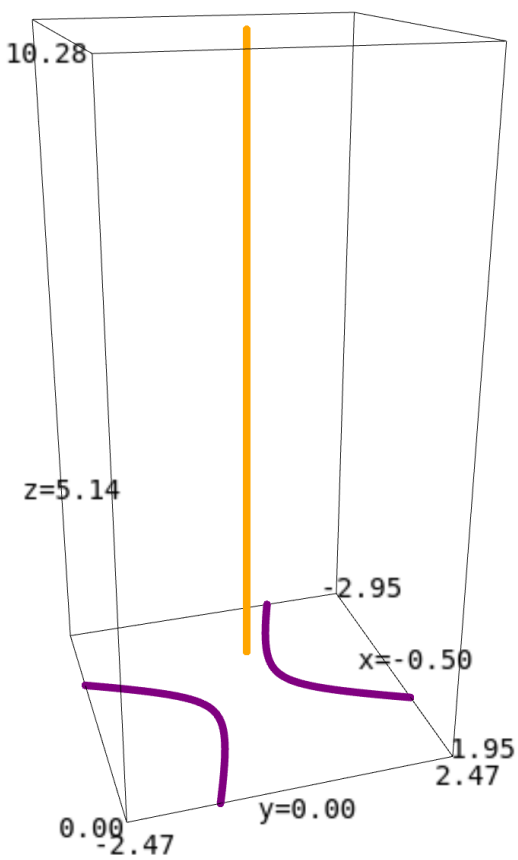


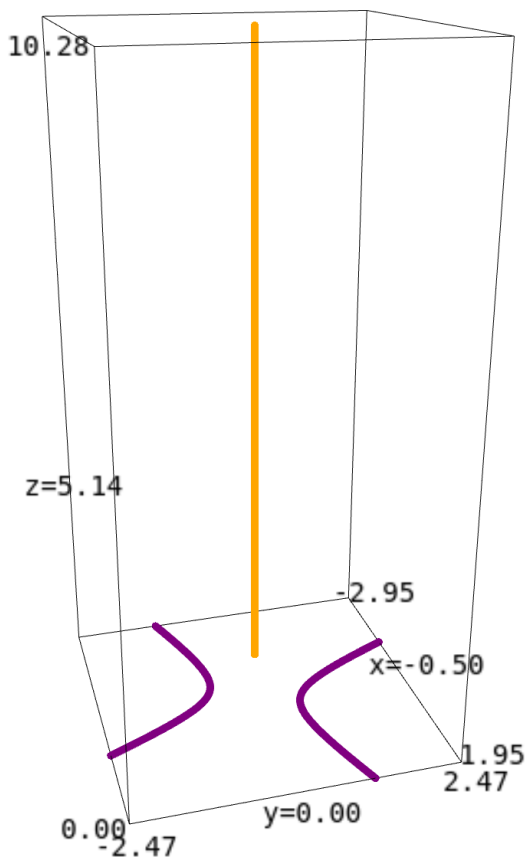
These two figures represent the result of varying b and β between -10 and 10 respectively.





These two figures represent the result of varying c and γ between -10 and 10 respectively.





In each case the purple lines are the paths of the roots, while orange is the vertex. This only begins to scratch the surface of the effects of varying parameters of the equations since each combination of variables does result in notably different behaviour. We would require 2^6 figures to capture the full scope of the possible movements. Without going to those lengths, there are at least some conclusions to be drawn from this exploration.

If we look back to (5) and (6), we can immediately see that c and γ will simply serve to translate one of the underlying conics. As a result, the roots will follow the path of the unaffected conic essentially tracing out a hyperbola as they move. While the roots follow their path, the xy position of the vertex remains unaffected; it does vary in z however.

The effect of b and β on the features of the quadratic modulus is of particular interest. The general effect is to move the roots linearly, in opposite directions, with b acting perpendicular to β. This movement causes the vertex to vary vertically rather vigorously.

Perhaps the most interesting behaviour is caused by variations in the a and α pair. This doesn't come as a surprise since squares of each are present in multiple denominators throughout the explicit formulas for both the roots and vertex. When adjusting one or the other, we witness the roots following a circular path tangent to the origin. The apogee of the path occurs when the value is closest to zero, and it approaches the origin asymptotically as the magnitude of the value increases.

One final note is that the real and imaginary components are mirror images of each other. These are rather elementary observations about the behaviour of the roots and vertex as the parameters are varied; a proper treatment would require a great deal more effort and space than is available in this short treatise on the subject. The more substantive material is in the derivation of explicit expressions for locations of the critical points of the equation generally.

4 Conclusion

With much effort, we were able to discover an explicit expression for the roots of the modulus of a quadratic equation over the field of complex numbers. A mundane quadratic expression in one variable and three coefficients, becomes nearly incomprehensible with the introduction of complex values. Despite the added difficulty, perhaps because of it, the satisfaction in finding an explicit analog to the old quadratic formula is tremendously pleasing. The field of complex numbers can seem dark and mysterious when first encountered. Despite obeying the field axioms, the exotic behaviour it leads to makes it seem like a very different object. This is illustrated perfectly when looking at the partially expanded explicit expression for the roots of a complex quadratic's modulus, especially when contrasted with the real variety.

A great deal more can be explored with respect to the specific changes resulting from varying parameters, but to have simply uncovered a tool to accomplish that end with greater ease is sufficient; a deeper understanding of the underlying principles may be necessary for more intricate analyses. There is likely insight to be gained by looking at each component of the critical values in isolation as it is affected by the changes in the coefficients. Building on the work presented here, an exhaustive survey of the effects of the parameters of a complex quadratic's modulus could be automated with ease.

A Workings in SageMath

Below is the code used to generate the different parametric plots of action of the roots and vertex in the SageMath environment. With only minor modification, it can be reverted to produce plots of the modulus function with roots and vertex marked out with spheres.

```
%display latex
var('a b c alpha beta gamma x y z', domain = 'real')
a=1
b=1
c=1
alpha=1
#beta=1
gamma=1
A=a+I*alpha
B=b+I*beta
C=c+I*gamma
Z=x+i*y
f=A*Z^2+B*Z+C
f_real=real(f)
f_imag=imag(f)
f_mod=sqrt(f_real^2+f_imag^2)
f_mod
x_v(beta)=-(a*b+alpha*beta)/(2*a^2+2*alpha^2)
y_v(beta)=(alpha*b-a*beta)/(2*a^2+2*alpha^2)
z_v(beta)=f_mod.subs({x:x_v,y:y_v})
p=x_v^2-y_v^2-((a*c+alpha*gamma)/(a^2+alpha^2))
q=x_v*y_v-(a*gamma-alpha*c)/(2*a^2+2*alpha^2)
y_root_1_origin=-sqrt((-p+sqrt(p^2+4*q^2))/2)
x_root_1_origin=q/y_root_1_origin
y_r1(beta)=y_root_1_origin+y_v
y_r2(beta)=y_v-y_root_1_origin
x_r1(beta)=x_root_1_origin+x_v
x_r2(beta)=x_v-x_root_1_origin
Roots = parametric_plot3d((x_r1(beta), y_r1(beta),0), (beta, -10, 10), color='purple') + \
parametric_plot3d((x_r2(beta), y_r2(beta),0), (beta, -10, 10), color='purple') + \
parametric_plot3d((x_v(beta) , y_v(beta), z_v(beta)), (beta, -10, 10), color='orange')
```

B A Restatement of the Equations

$$
\begin{aligned}
f(x,y) &= \sqrt{(ax^2 - 2\alpha xy - ay^2 + bx - \beta y + c)^2 + (\alpha x^2 + 2axy - \alpha y^2 + \beta x + by + \gamma)^2} \\
x_v &= -\frac{ab + \alpha\beta}{2a^2 + 2\alpha^2} \\
y_v &= \frac{\alpha b - a\beta}{2a^2 + 2\alpha^2} \\
z_v &= f(x_v, y_v) \\
p &= x_v^2 - y_v^2 - \frac{ac + \alpha\gamma}{a^2 + \alpha^2} \\
q &= x_v y_v - \frac{a\gamma - \alpha c}{2a^2 + 2\alpha^2} \\
y_{r1} &= y_v - \sqrt{\frac{-p + \sqrt{p^2 + 4q^2}}{2}} \\
y_{r2} &= y_v + \sqrt{\frac{-p + \sqrt{p^2 + 4q^2}}{2}} \\
x_{r1} &= x_v - \frac{q}{\sqrt{\frac{-p + \sqrt{p^2 + 4q^2}}{2}}} \\
x_{r2} &= x_v + \frac{q}{\sqrt{\frac{-p + \sqrt{p^2 + 4q^2}}{2}}}
\end{aligned}
$$

Acknowledgements

I would like to extend my deepest gratitude to ChatGPT for their tireless efforts proofreading my work and providing valuable suggestions.